

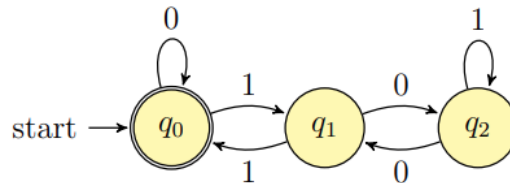
CSE331: Automata and Computability

Assignment 1

Deadline: 23/07/2026 11:59 PM

Question 1

Consider the finite automaton $(Q, \Sigma, \delta, q_0, F)$ depicted below



where $Q = \{q_0, q_1, q_2\}$, $\Sigma = \{0, 1\}$, and $F = \{q_0\}$.

- Find and fix the mistake in the following symbolic description of the transition function $\delta : Q \times \Sigma \rightarrow Q$: for each $j \in \{0, 1\}$
 $\delta(q_0, j) = q_j$ $\delta(q_1, j) = q_{1-j}$ $\delta(q_2, j) = q_{1+j}$
- Keeping the same set of states Q , alphabet Σ , starting state q_0 , and set of accepting states $\{q_0\}$, change the transition function δ so that the resulting finite automaton never accepts a string ending in a 1.
- There is a beautiful plain English description of the language recognized by the finite automaton with the state diagram depicted at the start of Question 1. What is it?

Question 2

For each of the languages below, draw the DFA that recognizes the language. You may assume that each language is over the alphabet $\Sigma = \{0, 1\}$.

- $\{w \in \Sigma^* \mid \text{the set of all string which do not end with } 11\}$
- $\{w \in \Sigma^* \mid \text{if the number of 1's in } w \text{ is odd, then the number of 0's in } w \text{ is also odd}\}$
- $\{w \in \Sigma^* \mid \text{binary strings that contain an even number of 1s and an odd number of 0's and do not contain the substring } 10.\}$ (*You need to construct this language without cross product*)

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Question 3

Let $\Sigma = \{a, b\}$. Consider the following languages over Σ .

$$L_1 = \{w : \text{length of } w \text{ is two more than multiple of four}\}$$

$$L_2 = \{w : \text{every even position letter in } w \text{ is different from the first letter of } w\}$$

$$L_3 = \{w : \text{every } 2k + 1 \text{ position in } w \text{ is } b, \text{ where } k \geq 0\}$$

$$L_4 = \{w : \text{every } 2k + 1 \text{ position in } w \text{ is } a, \text{ where } k \geq 0\}$$

Now solve the following problems.

- (a) **Give** the state diagram for a DFA that recognizes L_1 . (3 points)
- (b) **Give** the state diagram for a DFA that recognizes L_2 . (3 points)
- (c) **Give** the state diagram for a DFA that recognizes L_3 . (3 points)
- (d) If you were to use the “cross product” construction to obtain a DFA for the language $L_2 \cap (L_3 \cup L_4)$, how many states would it have? (1 point)
- (e) **Find** all four-letter strings in $L_2 \cap (L_3 \cup L_4)$. (1 point)
- (f) **Give** the state diagram for a DFA that recognizes $L_2 \cap (L_3 \cup L_4)$ using only four states. (2 points)
- (g) **Find** a four-letter string in $\overline{L_3} \circ L_4$. [Recall: $\overline{L}$ denotes the complement of the language L i.e., $\overline{L} = \Sigma^* - L$] (1 point)
- (h) Is $\overline{L_3} \circ L_4 = \overline{L_3}$? **Give** justification for your answer. (1 point)

Question 4

For each of the languages below, write the regular expression for the given language. You may assume that each language is over the alphabet $\Sigma = \{0, 1\}$.

- (a) $\{w \in \Sigma^* \mid w \text{ contains equal number of } 01 \text{ and } 10\}$
- (b) $\{w \in \Sigma^* \mid w \text{ do not contain substring } ba\}$
- (c) $\{w \in \Sigma^* \mid w \text{ is the set of all strings whose } 1\text{st}, 3\text{rd}, 5\text{th}, \dots, \text{ characters are the same}\}$
- (d) $\{w \in \Sigma^* \mid w \text{ starts with } 0 \text{ and has odd length, or starts with } 1 \text{ and has even length}\}$

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Question 5

Let $\Sigma = \{0, 1\}$. Consider the following languages over Σ .

$$L_1 = \{w \text{ contains exactly two } 1\}$$

$$L_2 = \{w \text{ doesn't start with } 0\}$$

$$L_3 = \{\text{every third position in } w \text{ is } 1\}$$

$$L_4 = \{\text{every } 1 \text{ in } w \text{ is followed by at least two } 0\}$$

$$L_5 = L_3 \cap L_4$$

Now solve the following problems.

- (a) **Give** a regular expression for the language L_1 . (3 points)
- (b) **Give** a regular expression for the language L_2 . (3 points)
- (c) **Give** a regular expression for the language L_3 . (3 points)
- (d) **Write** a five-letter string that belongs to L_5 . (1 point)
- (e) **Give** a regular expression for the language L_5 . (2 points)
- (f) **Give** a regular expression for the language $\bar{L}_4$. Here $\bar{L}$ denotes the complement of the language L i.e., $\bar{L} = \Sigma^* - L$. (3 points)

Question 6

Convert the following DFA into an equivalent regular expression using the state elimination method. First eliminate q_2 , then q_1 , and finally q_0 . You must show work.

